

Home Buying RAG

15-question evaluation

A source-level assessment of retrieval precision, recall, ranking quality, and observed failure patterns across the clean 33-document reference corpus.

60% STRICT PASS RATE	67% HIT@1	93% HIT@3	100% HIT@5	0.806 MEAN RECIPROCAL RANK
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EVALUATED	CORPUS	VECTOR BASELINE
August 24, 2026 at 05:51 UTC	33 documents / 604 pages	2,776 clean chunks

Overall assessment

Retrieval shows strong broad recall but moderate rank-one precision. Every test found an expected source within five unique sources, while five questions placed a different relevant or adjacent source first. The system is suitable for a multi-source RAG answer flow, but the first result should not be treated as independently authoritative.

Scope: judgments are binary at the source level. This report does not grade final answer writing, citation correctness, or whether each retrieved chunk contains every required fact.

Question-level results

Expected sources were selected manually from the curated corpus. Rank is measured after deduplicating returned chunks by source document.

ID	Verdict	Category	Expected source	Rank	Hit@5	Recall	Top score
Q0 1	FAIL	Closing costs	cfpb_home_loan_toolkit.pdf cfpb_closing_disclosure_sample.pdf	2	Yes	50%	0.743
Q0 2	PASS	Adjustable-rate mortgages	cfpb_arm_charm_booklet.pdf	1	Yes	100%	0.957
Q0 3	FAIL	Loan estimates	cfpb_loan_estimate_sample.pdf cfpb_home_loan_toolkit.pdf	1	Yes	50%	0.971
Q0 4	PASS	California disclosures	ca_civil_code_1102_disclosure_law.html ca_dre_refbook_disclosures.pdf	1	Yes	100%	0.866
Q0 5	PASS	Texas disclosures	trec_sellers_disclosure_notice.pdf	1	Yes	100%	0.932
Q0 6	FAIL	Homeowners associations	trec_hoa_addendum.pdf	4	Yes	100%	0.322
Q0 7	FAIL	Home inspections	ashi_home_inspection_standards.pdf	2	Yes	100%	0.986
Q0 8	PASS	Federal income taxes	irs_pub530_homeowners_tax.pdf	1	Yes	100%	0.757
Q0 9	FAIL	Appraisals	fannie_mae_appraisal_report_1004.pdf	2	Yes	100%	0.670
Q1 0	PASS	Homeowners insurance	naic_homeowners_insurance_guide.pdf	1	Yes	100%	0.711
Q1 1	FAIL	Fair housing	hud_fair_housing_booklet.pdf	3	Yes	100%	0.932
Q1 2	PASS	Contractors	ftc_hiring_a_contractor.pdf	1	Yes	100%	0.713
Q1 3	PASS	VA loans	va_home_loan_buyers_guide.pdf	1	Yes	100%	0.869
Q1 4	PASS	USDA loans	usda_rural_housing_loan_guide.pdf	1	Yes	100%	0.866
Q1 5	PASS	California down payment help	calhfa_downpayment_assistance.html	1	Yes	100%	0.999

Metric definitions

- Strict pass: an expected source ranks first and all expected sources appear within the top five.
- Hit@K: at least one expected source appears within the first K unique sources.
- MRR: rewards placing the first expected source near rank one; 1.0 is perfect.
- Source recall@5: fraction of all expected sources recovered in the top five.
- Reranker scores are comparative relevance signals, not calibrated probabilities.

Failure analysis

Strict failures

6 of 15 questions failed the strict criterion. A failure means the expected source was not ranked first or at least one manually expected source was absent from the top five.

- Q01 / Closing costs: FAIL - expected source not ranked first; incomplete expected-source recovery.
- Q03 / Loan estimates: FAIL - incomplete expected-source recovery.
- Q06 / Homeowners associations: FAIL - expected source not ranked first.
- Q07 / Home inspections: FAIL - expected source not ranked first.
- Q09 / Appraisals: FAIL - expected source not ranked first.
- Q11 / Fair housing: FAIL - expected source not ranked first.

Expected source retrieved, but not ranked first

- Q01 / Closing costs: expected source first appeared at rank 2; rank one was `va_home_loan_buyers_guide.pdf`.
- Q06 / Homeowners associations: expected source first appeared at rank 4; rank one was `trec_new_home_contract_incomplete.pdf`.
- Q07 / Home inspections: expected source first appeared at rank 2; rank one was `va_home_loan_buyers_guide.pdf`.
- Q09 / Appraisals: expected source first appeared at rank 2; rank one was `va_home_loan_buyers_guide.pdf`.
- Q11 / Fair housing: expected source first appeared at rank 3; rank one was `cfpb_home_loan_toolkit.pdf`.

Partial multi-source recovery

- Q01 / Closing costs: missing `cfpb_closing_disclosure_sample.pdf` from the top five.
- Q03 / Loan estimates: missing `cfpb_loan_estimate_sample.pdf` from the top five.

Cross-cutting observations

- Forms and sample disclosures may retrieve boilerplate instead of the exact field needed.
- Jurisdiction terms are essential; vague questions may blend California, Texas, and federal sources.
- Broad guides can outrank specialist documents because they cover many buyer topics in natural language.
- Source relevance does not guarantee complete answer coverage at the chunk level.

Recommended actions

- Preserve clean namespace rebuilds so obsolete chunk IDs do not accumulate.
- Add title, publisher, jurisdiction, category, and source URL metadata during ingestion.
- Introduce jurisdiction-aware filters or routing for state-specific questions.
- Add chunk-level relevance judgments for ambiguous and form-based questions.
- Track this suite after every corpus, embedding, chunking, or reranker change.

Retrieved-source detail

The first five unique source documents returned for each evaluation question.

Q01 / What costs and fees should a buyer expect to see at closing?

1	va_home_loan_buyers_guide.pdf	Page 12.0	0.743
2	cfpb_home_loan_toolkit.pdf	Page 21.0	0.593
3	irs_pub530_homeowners_tax.pdf	Page 14.0	0.590
4	trec_condo_resale_contract.pdf	Page 4.0	0.563
5	hud_home_buying_guide.pdf	Page 6.0	0.377

Q02 / How can the interest rate and monthly payment change on an adjustable-rate mortgage?

1	cfpb_arm_charm_booklet.pdf	Page 1.0	0.957
2	hud_home_buying_guide.pdf	Page 5.0	0.903
3	ca_dre_refbook_financing.pdf	Page 6.0	0.615

Q03 / How should I use a Loan Estimate to compare mortgage offers from different lenders?

1	cfpb_home_loan_toolkit.pdf	Page 11.0	0.971
2	cfpb_arm_charm_booklet.pdf	Page 1.0	0.897

Q04 / What defects and conditions must a California home seller disclose to a buyer?

1	ca_dre_refbook_disclosures.pdf	Page 17.0	0.866
2	ca_civil_code_1102_disclosure_law.html	Page unknown	0.750
3	ca_dre_refbook_agency.pdf	Page 39.0	0.741
4	trec_purchase_contract.pdf	Page 3.0	0.341

Q05 / What information is included in a Texas seller's disclosure notice?

1	trec_sellers_disclosure_notice.pdf	Page 0.0	0.932
2	trec_purchase_contract.pdf	Page 3.0	0.839
3	trec_condo_resale_contract.pdf	Page 3.0	0.796
4	trec_new_home_contract_completed.pdf	Page 3.0	0.329

Q06 / What HOA documents, fees, and restrictions should a Texas buyer review?

1	trec_new_home_contract_incomplete.pdf	Page 4	0.322
2	trec_purchase_contract.pdf	Page 4.0	0.286
3	trec_new_home_contract_completed.pdf	Page 4.0	0.219
4	trec_hoa_addendum.pdf	Page 0.0	0.127
5	trec_condo_resale_contract.pdf	Page 5.0	0.125

Q07 / What systems and components should a standard home inspection cover?

1	va_home_loan_buyers_guide.pdf	Page 25.0	0.986
2	ashi_home_inspection_standards.pdf	Page 8.0	0.972

Q08 / Which homeownership expenses may be deductible on my federal income tax return?

1	irs_pub530_homeowners_tax.pdf	Page 2.0	0.757
2	cfpb_closing_disclosure_sample.pdf	Page 12.0	0.052

Q09 / What property details and comparable sales are documented in a residential appraisal report?

1	va_home_loan_buyers_guide.pdf	Page 24.0	0.670
2	fannie_mae_appraisal_report_1004.pdf	Page 0.0	0.390
3	tx_property_tax_basics.pdf	Page 10.0	0.246
4	ca_dre_refbook_financing.pdf	Page 103	0.217

Q10 / What should I compare when choosing a homeowners insurance policy?

1	naic_homeowners_insurance_guide.pdf	Page 13.0	0.711
2	cfpb_home_loan_toolkit.pdf	Page 18.0	0.215

Q11 / What kinds of discrimination are prohibited when buying or financing a home?

1	cfpb_home_loan_toolkit.pdf	Page 16.0	0.932
2	ca_dre_refbook_financing.pdf	Page 77.0	0.796
3	hud_fair_housing_booklet.pdf	Page 4.0	0.683
4	ca_dre_refbook_agency.pdf	Page 33.0	0.668

Q12 / What should I verify before hiring a contractor for repairs or renovations?

1	ftc_hiring_a_contractor.pdf	Page 2.0	0.713
2	ca_civil_code_1102_disclosure_law.html	Page unknown	0.046
3	va_home_loan_buyers_guide.pdf	Page 16.0	0.030

Q13 / What are the main eligibility, funding fee, and occupancy considerations for a VA home loan?

1	va_home_loan_buyers_guide.pdf	Page 6.0	0.869
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Q14 / Who may qualify for a USDA rural housing loan and what properties are eligible?

1	usda_rural_housing_loan_guide.pdf	Page 1.0	0.866
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Q15 / What down payment assistance options does CalHFA offer California home buyers?

1	calhfa_downpayment_assistance.html	Page unknown	0.999
2	ca_dre_refbook_financing.pdf	Page 171.0	0.924